

Akash Kumar Singh

Full Stack Developer · AI/ML & Prompt Engineering · MCA Student

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SUMMARY

Full-stack developer and MCA student (AI & ML) at Amity University Online. Builds real-time web applications and prompt-engineering tools — from Kafka-powered location trackers to OWASP LLM01 red-teaming platforms — with all projects deployed and accessible online. Deepening expertise in distributed systems, generative AI, and modern web architecture. Actively seeking internships, freelance work, and entry-level full-time opportunities.

TECHNICAL SKILLS

Frontend	React, Next.js, TypeScript, Tailwind CSS, Framer Motion
Backend	Node.js, Express, FastAPI, Apache Kafka, WebSockets, REST APIs
Databases	PostgreSQL, Redis, SQLite, MongoDB
Languages	JavaScript, TypeScript, Python
AI / ML & Prompt	NumPy, core ML algorithms, LLM API integration, Anthropic Claude, Prompt Engineering
Tools	Git, GitHub, Vercel, Turborepo, tRPC, Docker (learning) · React Native / Flutter (exploring)

PROJECTS

ChaiForm — Form Builder SaaS

Next.js · TypeScript · Express · PostgreSQL · Turborepo · tRPC · Drizzle ORM · Zod | [chaiforms-web-hjco.onrender.com](#) | [github.com/Akashkr28/ChaiForm_Form_Builder_SaaS](#)

- Production-grade Typeform-inspired form builder on a Turborepo monorepo — Google OAuth, dynamic field schemas, response analytics, CSV export, type-safe tRPC API with Drizzle ORM & Zod, email notifications, and honeypot spam protection.

Pulseboard — Live Polling & Feedback Platform

React · TypeScript · Node.js · Socket.io | [pulse-board-live-poll-for-feedback.vercel.app](#) | [github.com/Akashkr28/PulseBoard-Live-Poll-for-Feedback](#)

- Real-time collaborative polling platform; instant vote-count updates broadcast to all connected clients via WebSocket with shareable links and live result visualization.

1 Million Checkboxes — Collaborative Real-time App

React · Node.js · Redis | [million-checkboxes-xtg8.onrender.com](#) | [github.com/Akashkr28/1-million-checkboxes](#)

- Synchronized 1,000,000 shared checkboxes across all concurrent users in real-time via WebSocket broadcast with an optimized rendering pipeline for consistent DOM performance.

Real-time Location Tracker — Kafka-powered GPS System

React · Apache Kafka · Node.js · Leaflet.js | [github.com/Akashkr28/location-tracker](#)

- Streamed live GPS coordinates through Kafka topics for high-throughput, fault-tolerant event delivery; rendered sub-second map updates using Leaflet.js on the consumer side.

Prompt Engineering Projects

Sentinel — Prompt Injection & Attack Defense

Next.js · TypeScript · Python · SQLite · Anthropic Claude API | [sentinel-prompt-injection.vercel.app](#) | [github.com/Akashkr28/Sentinel_Prompt_Injection_Attack_Defense](#)

- Full-stack OWASP LLM01 red-teaming platform with dual-layer detection — heuristic regex (<1 ms) and LLM-as-judge semantic grading — benchmarking 21 payloads across 5 injection categories with 8 benign controls for honest false-positive measurement.

PromptForge — Automated Prompt Optimizer

Next.js · TypeScript · Python · FastAPI · SQLite · Anthropic Claude API | [prompt-forge-a-prompt-optimizer-too.vercel.app](#) | [github.com/Akashkr28/Prompt_Forge_A_Prompt_Optimizer_Tool](#)

- Self-improving prompt engine that iterates execute → evaluate → optimize loops via meta-prompting and concurrent LLM sampling, tracking gains through live score curves, word-level diffs, and searchable session history.

EDUCATION

Master of Computer Applications (MCA) — AI & ML Specialization

Amity University Online · 2023 – Present · 3rd Semester (Current)

- Coursework: Machine Learning, Data Structures, Distributed Systems, Python for AI; applied through production-grade real-time and AI-driven projects deployed throughout the programme.

ADDITIONAL

Interests: Open-source development, real-time systems architecture, generative AI, prompt engineering, mobile-first UX. **Languages:** English (professional), Hindi (native). **Availability:** Immediately available for internships, freelance projects, and entry-level full-time roles.